

Experiment 'p_6_2_j_has_duplicates_seq' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_2_j_has_duplicates_seq

Natural language statement: Write a method to return true if the given array contains duplicate elements (which need not be adjacent).

Method signature: p_6_2_j_has_duplicates_seq(inputs: seq<int>) returns (has_duplicates: bool)

Ensures

- `has_duplicates == (exists i, j :: 0 <= i < j < |inputs| && inputs[i] == inputs[j])`

Requires

- `|inputs| >= 2`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to return true if the given array contains duplicate elements (which need
→ not be adjacent).

The signature should be:

p_6_2_j_has_duplicates_seq(inputs: seq<int>) returns (has_duplicates: bool)

The method should respect the following contract:

requires |inputs| >= 2, ensures has_duplicates == (exists i, j :: 0 <= i < j < |inputs|
→ && inputs[i] == inputs[j])

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn1T3WBuz951GnbIPA4EBx6sFOcoX

Model: gpt-5.5-2026-04-23

Created at: 1780576085

Finish reason: stop

Usage: 330 tokens in, and 295 tokens out

```
{
  has_duplicates := exists i, j :: 0 <= i < j < |inputs| && inputs[i] == inputs[j];
}
```

Final Program

```
method p_6_2_j_has_duplicates_seq(inputs: seq<int>) returns (has_duplicates: bool)
  requires |inputs| >= 2
  ensures has_duplicates == (exists i, j :: 0 <= i < j < |inputs| && inputs[i] ==
    → inputs[j])
{
  has_duplicates := exists i, j :: 0 <= i < j < |inputs| && inputs[i] == inputs[j];
}
```

Total Token Usage

Input tokens: 330

Output tokens: 295

Reasoning tokens: 246

Sum of ‘total tokens’: 625

Experiment Timings

Overall Experiment started at 1780576085393, ended at 1780576120438, lasting 35045ms (35.05 seconds)

Iteration #1 started at 1780576085399, ended at 1780576120437, lasting 35038ms (35.04 seconds)